DeepTrace++: A Graph Neural Network Approach for Epidemic Source Detection

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***Abstract*—Digital contact tracing and epidemic source detection are critical for controlling infectious disease outbreaks. Recent advances have modeled this problem as a graph-based inference task, where the objective is to identify the most likely source of contagion from partial observations of infection spread. Existing approaches, including centrality-based heuristics and message-passing algorithms, suffer from scalability issues and require prior knowledge of epidemic dynamics. Graph Neural Networks (GNNs) have emerged as a promising alternative, enabling data-driven learning of source likelihoods directly from network structures. However, current methods rely on static exploration strategies and limited feature representations.**

In this work, we propose DeepTrace++, an extension of the DeepTrace framework that introduces (i) centrality-aware node features, (ii) edge-weighted message passing, and (iii) a GNN-guided adaptive forward tracing strategy. By integrating structural and interaction-level information into the learning process and replacing heuristic graph traversal with a learned top-k exploration policy, the proposed approach aims to improve both efficiency and accuracy of source detection. Experimental evaluation on synthetic epidemic networks demonstrates the effectiveness of the proposed enhancements in identifying super-spreaders under realistic conditions.

***Index Terms*—Epidemic Source Detection, Contact Tracing, Graph Neural Networks, Message Passing, Susceptible-Infected (SI) Model, Network Diffusion, Centrality Measures, Weighted Graphs, Adaptive Exploration, Top-k Node Ranking**

I. Introduction

The rapid spread of infectious diseases creates an urgent need for efficient methods to identify infection sources and trace transmission paths. In epidemic networks, locating the first infected individual, often referred to as the source or patient zero, is important because it helps public health systems understand how the outbreak began, which individuals are likely to be exposed next, and where intervention should be focused. The source detection problem is not limited to biological epidemics; it also appears in misinformation spread, malware propagation, and other diffusion processes on graphs. Because of this broad relevance, source detection has become a central problem in graph-based inference and network science.

Traditional source detection methods rely on graph-theoretic measures such as rumor centrality, Jordan center, belief propagation, and dynamic message passing. These methods can work well in simplified settings, especially on trees or when the spreading dynamics are fully known, but they often struggle on general graphs with cycles, heterogeneous degrees, and limited observations. In particular, many classical approaches require knowledge of epidemic parameters or a precise observation time, which is unrealistic in practical outbreak scenarios. As the infected subgraph grows, the number of possible transmission histories also grows rapidly, making exact maximum-likelihood inference computationally expensive.

Recent work has shown that Graph Neural Networks (GNNs) provide a promising alternative for source detection because they can learn structural patterns directly from simulated epidemic data. Instead of explicitly computing transmission probabilities for every possible source node, a GNN can learn to rank nodes by their likelihood of being the origin of the outbreak. Studies such as Shah et al. and the DeepTrace paper show that GNN-based source detection can outperform classical methods in both accuracy and runtime, while also being more flexible across graph types. DeepTrace further frames the tracing process as an online graph exploration problem and combines forward and backward tracing with maximum-likelihood estimation on epidemic subgraphs.

However, existing DeepTrace-style systems still leave room for improvement. The original framework uses fixed exploration strategies such as BFS or DFS, and its node features are limited to a small set of handcrafted structural signals. It also treats graph edges uniformly during message passing, even though real-world contacts are often heterogeneous in strength and importance. These limitations motivate a more adaptive and information-rich extension of the framework.

In this paper, we therefore extend DeepTrace into a stronger framework, referred to here as **DeepTrace++**, by introducing richer node features, edge-weighted message passing, and adaptive forward tracing based on ranked node likelihoods. The central idea is to preserve the original maximum-

likelihood tracing objective while improving the model's ability to prioritize informative nodes and more realistic transmission pathways. This makes the overall contact tracing process more efficient, more flexible, and better suited to heterogeneous epidemic networks.

II. Background

A. Source Detection in Epidemic Networks

The source detection problem asks: given a graph representing contacts and a partially observed epidemic snapshot, which node most likely started the outbreak? In network terms, the graph $G = (V, E)$ contains individuals as nodes and interactions as edges. The observed infected subgraph is only a partial view of the true epidemic network, and the task is to infer the source node or the set of source nodes that best explains the observation. This is typically formulated as a maximum-likelihood estimation problem, where the goal is to find the source candidate that maximizes the probability of generating the observed infection pattern.

In the DeepTrace formulation, the contact tracing process is split into two parts: **forward tracing** and **backward tracing**. Forward tracing grows the observed subgraph from an index case by exploring neighboring contacts, while backward tracing estimates which node in the currently observed graph is most likely to be the superspreader. The observed contact tracing network is denoted as G_n, and the estimator seeks the node v that maximizes $P(G_n \mid v)$. This makes source detection a sequential inference problem over an incrementally revealed graph.

B. The SI Spreading Model

For this project, the spreading process is modeled using the **Susceptible-Infected (SI)** model, which is one of the simplest and most widely used epidemic spreading models in source detection studies. Under the SI model, each node is in one of two states: susceptible or infected. Once a node becomes infected, it remains infected for the rest of the process. Infection spreads from infected nodes to neighboring susceptible nodes along graph edges. The DeepTrace paper focuses on the SI model because it is suitable for early-stage outbreaks, where recovery is not yet dominant and the key problem is identifying the initial spreader from a growing infected subgraph.

In the SI setting, the likelihood of an observed infected subgraph depends on the infection order and on the structure of the underlying graph. For a given candidate source node, there can be many valid infection sequences, and the likelihood is obtained by considering all permitted permutations consistent with the observed graph. This is what makes exact inference difficult on large or cyclic graphs. The number of possible infection histories grows quickly with graph size, which is why classical exact maximum-likelihood estimation becomes computationally expensive.

C. Problem Formulation

Given an observed contact tracing graph $G_n \subseteq G$, the goal is to estimate the most likely superspreader:

$$\hat{v} \in \arg\max_{v \in G_n} P(G_n \mid v) \tag{1}$$

Here, $P(G_n \mid v)$ is the likelihood that candidate node v generated the observed tracing graph under the SI spreading process. In the full DeepTrace formulation, this likelihood can be expressed through the set of permitted infection permutations rooted at v. The backward tracing objective is therefore to find the node whose infection history best matches the observed epidemic subgraph.

This formulation is tightly coupled to the forward tracing process. As more nodes are discovered, the observed graph grows, the likelihood scores change, and the current best estimate of the source may shift. In the original DeepTrace framework, this is handled through iterative tracing using BFS or DFS expansion of the infection subgraph. The project therefore becomes an online graph exploration problem in which the tracer must balance coverage, efficiency, and inference quality.

D. Why Graph Neural Networks are Used

GNNs are well suited to source detection because the source node is not identified by its state alone, but by the structure of infected and uninfected neighbors around it. GNNs learn node embeddings by repeatedly aggregating local neighborhood information, which naturally matches the structure of epidemic diffusion. Prior work shows that GNN-based source detection can learn meaningful representations of infection patterns and outperform classical methods on synthetic and realistic graphs. DeepTrace builds on this idea by training a GNN to approximate the likelihood $P(G_n \mid v)$ for each candidate node.

The DeepTrace paper also shows that the method can be trained on synthetic epidemic networks and then fine-tuned using higher-quality labels derived from exact or approximate maximum-likelihood computations. This two-phase training strategy helps the model learn the source-detection task more effectively while reducing the cost of obtaining training labels.

E. Motivation for the Proposed Extension

Although DeepTrace already combines graph exploration and likelihood estimation, it still uses limited node descriptors and fixed forward-tracing behavior. Existing literature also shows that feature augmentation, temporal observation structure, and source-detection outputs can significantly influence performance, but the effect of richer graph features and adaptive exploration has not been fully explored in a unified framework. This creates a clear opportunity for improvement through centrality-aware inputs, weighted aggregation, and ranked top-k expansion.

Accordingly, the proposed DeepTrace++ framework keeps the same SI-based source detection objective but extends the model with stronger structural priors and a more adaptive exploration policy. This preserves compatibility with the original

problem formulation while making the tracing process more expressive and potentially more efficient.

III. DeepTrace Algorithm

DeepTrace formulates epidemic source detection as a maximum-likelihood inference problem over an incrementally observed graph, where the goal is to identify the most likely source node given partial observations of an epidemic spreading process. The algorithm operates on a dynamically growing contact tracing graph $G_n = (V_n, E_n)$, where $n = |V_n|$ denotes the number of observed infected nodes at a given stage of tracing.

A. Epidemic Model (SI Model)

DeepTrace assumes a **Susceptible-Infected (SI)** spreading process. Each node $v \in V$ has a state:

$$x_v(t) \in \{S, I\} \tag{2}$$

The infection spreads along edges such that if node u is infected, it can infect a neighboring node $v \in \mathcal{N}(u)$. Under a discrete-time formulation, the infection rule can be written as:

$$P(x_v(t+1) = I \mid x_v(t) = S) = 1 - \prod_{u \in \mathcal{N}(v)} \left(1 - \beta \cdot \mathbf{1}_{x_u(t)=I}\right) \tag{3}$$

where:

- β is the infection probability,
- $\mathbf{1}_{x_u(t)=I}$ indicates whether neighbor u is infected.

In the SI model, once a node becomes infected, it remains infected:

$$x_v(t) = I \Rightarrow x_v(t+1) = I \tag{4}$$

This assumption simplifies the inference problem and makes it suitable for early-stage outbreak modeling.

B. Problem Formulation (MLE Objective)

Given the observed contact tracing graph G_n, the goal is to estimate the source node:

$$\hat{v} = \arg\max_{v \in V_n} P(G_n \mid v) \tag{5}$$

Here, $P(G_n \mid v)$ denotes the likelihood that node v generated the observed infection subgraph under the SI process.

C. Likelihood via Permitted Infection Sequences

Under the SI model, the infection spreads sequentially, and any valid explanation of G_n corresponds to a permitted permutation (infection order) of nodes:

$$\pi = (v_1, v_2, \ldots, v_n) \tag{6}$$

such that:

- $v_1 = v$ (candidate source),
- each node v_i is connected to at least one previously infected node:

$$\exists j < i \text{ such that } (v_j, v_i) \in E_n \tag{7}$$

Let $\Pi(v)$ denote the set of all such valid permutations rooted at v. The likelihood is then:

$$P(G_n \mid v) = \sum_{\pi \in \Pi(v)} P(\pi) \tag{8}$$

1) Permutation Probability: The probability of a permutation π can be expressed as:

$$P(\pi) = \prod_{i=2}^{n} \frac{1}{|\mathcal{B}_{i-1}|} \tag{9}$$

where $\mathcal{B}_{i-1}$ is the boundary set of infected nodes at step $i-1$.

D. Forward Tracing (Graph Expansion)

DeepTrace constructs G_n incrementally starting from an index case v_0:

$$G_1 = \{v_0\} \tag{10}$$

At each step:

$$G_{n+1} = G_n \cup \mathcal{N}(v) \tag{11}$$

where v is selected using a traversal strategy.

Two strategies are analyzed:

1) BFS Expansion: Nodes are expanded in increasing distance from v_0.

2) DFS Expansion: Nodes are expanded along a path before backtracking.

E. Node Feature Representation

Each node $v \in V_n$ is represented using a feature vector:

$$h_v^{(0)} = [1, \hat{r}(v), \check{r}(v)]^T \tag{12}$$

1) Infected Ratio:

$$\hat{r}(v) = \frac{|\{u \in \mathcal{N}(v) : u \in V_n\}|}{|\mathcal{N}(v)|} \tag{13}$$

2) Boundary Distance Ratio:

$$\check{r}(v) = \frac{d(v, \partial G_n)}{\max_{u \in V_n} d(u, \partial G_n)} \tag{14}$$

F. Graph Neural Network (GraphSAGE)

The proposed GraphSAGE architecture consists of three message-passing layers with LSTM-based neighborhood aggregation. Each layer transforms node representations using learnable weight matrices followed by nonlinear activation functions. The hidden embedding dimension is set to 64.

Formally, the layer-wise update rule is:

$$h_v^{(l+1)} = \sigma\left(W^{(l)} \cdot \text{LSTM}\left(\{h_u^{(l)} : u \in \mathcal{N}(v)\}\right)\right) \tag{15}$$

After the final layer, node embeddings are passed through a fully connected prediction layer:

$$\hat{y}_v = W_o h_v^{(L)} + b_o \tag{16}$$

which outputs an estimate of the log-likelihood score for each candidate source node..

1) Message Passing Layer:

$$h_v^{(l+1)} = \sigma\left(W^{(l)} \cdot \text{AGG}\left(\{h_u^{(l)} : u \in \mathcal{N}(v)\}\right)\right) \tag{17}$$

For LSTM-based aggregation:

$$\text{AGG} = \text{LSTM}\left(\{h_u^{(l)}\}\right) \tag{18}$$

2) Final Prediction:

$$\hat{y}_v = f(h_v^{(L)}) \tag{19}$$

where $\hat{y}_v \approx \log P(G_n \mid v)$.

G. Learning Objective

The model is trained using supervised learning:

$$\mathcal{L} = \sum_{v \in V_n} (\hat{y}_v - y_v)^2 \tag{20}$$

where y_v is the (approximate or exact) log-likelihood.

H. Two-Phase Training

1) Phase 1: Pre-training:

$$y_v^{\text{approx}} \approx \log P(G_n \mid v) \tag{21}$$

Estimated via sampling permutations.

2) Phase 2: Fine-tuning:

$$y_v^{\text{exact}} = \log P(G_n \mid v) \tag{22}$$

Computed for special graph classes such as trees.

I. Overall Algorithm (Iterative Process)

At each iteration t:

1) Observe graph G_t,
2) Compute node features $h_v^{(0)}$,
3) Run GNN:

$$\hat{y}_v = \text{GNN}(G_t) \tag{23}$$

4) Estimate source:

$$\hat{v}_t = \arg\max_v \hat{y}_v \tag{24}$$

5) Expand graph using BFS/DFS,
6) Repeat.

J. Complexity Insight

Exact MLE is exponential in n, whereas DeepTrace with GNN has complexity:

$$\mathcal{O}(L \cdot |E_n|) \tag{25}$$

which is significantly more scalable.

IV. Research Gaps

Although recent studies have made important progress in epidemic source detection using graph-theoretic methods and GNNs, several important gaps still remain. These gaps motivate the proposed DeepTrace++ framework and help justify why the existing DeepTrace pipeline can be extended further.

A. Limited Exploration Strategy in Forward Tracing

A major limitation of DeepTrace is that its forward tracing mechanism is built around fixed graph traversal strategies such as BFS and DFS. While these strategies are simple and analytically convenient, they do not adapt to the evolving likelihood scores produced by the model during tracing. As a result, the algorithm may expand less informative regions of the graph even when more promising candidate nodes are available elsewhere. The original DeepTrace paper itself notes that BFS and DFS behave differently depending on graph diameter, topology, and the position of the index case, which shows that no single static traversal strategy is universally optimal.

This creates a gap between **graph exploration** and **source inference**. In an ideal system, the tracing policy should be guided by the current belief about where the source is likely to be, rather than by a predetermined search order. This motivates an adaptive forward tracing rule, such as top-k expansion based on predicted likelihoods, which is not present in the original framework.

B. Insufficient Use of Rich Node-Level Structural Information

Existing GNN-based source detection methods do use node features, but in many cases these features are limited or only partially analyzed. In the DeepTrace paper, the initial features include the infected proportion and boundary distance ratio, which are useful but still relatively local descriptors of the infection pattern. The review literature also notes that several prior studies rely heavily on feature augmentation, but often do not provide a quantitative analysis of how much those features actually improve performance.

This leaves an open gap in feature design: current models may not sufficiently capture both **local infection structure** and **global graph position**. Important structural signals such as closeness centrality or distance from the index case are not consistently included, even though they can help the model understand how centrally located or how early-placed a node is within the growing epidemic subgraph. This makes feature enrichment a natural and still underexplored direction.

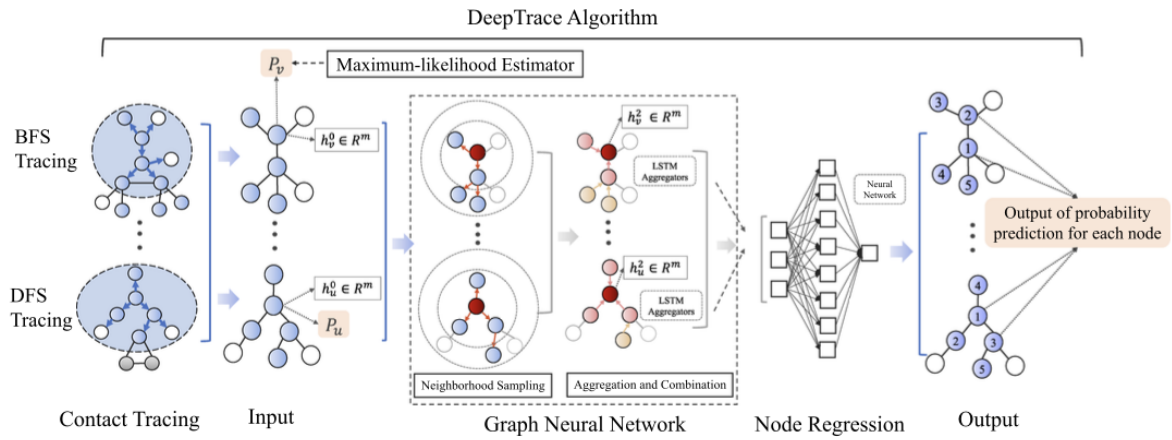

Fig. 1. The overall architecture of Algorithm DeepTrace employs a GNN, taking as input several small-scale networks obtained through BFS and DFS methods. Each node in these networks has structural features and a training label representing the permitted permutation probability. Semi-supervised learning is conducted using GraphSage with LSTM aggregators.

C. Edges Are Usually Treated as Uniform, Which Is Unrealistic

Most source detection methods, including the DeepTrace framework, effectively treat graph edges as equal during message passing and tracing. This is a strong simplification because real contact networks are heterogeneous: some contacts are brief, weak, or low-risk, while others are repeated, close, or high-risk. The DeepTrace paper discusses graph exploration and message passing, but it does not model varying interaction strengths directly in its core tracing mechanism.

This is an important research gap because edge heterogeneity can influence both the probability of transmission and the reliability of a contact as evidence. If all edges are treated identically, the model may overemphasize weak contacts and underweight stronger transmission pathways. A weighted message-passing mechanism can address this limitation by allowing the model to distinguish between more and less informative interactions.

D. Weak Coupling Between Forward Tracing and Backward Inference

Another gap is that many systems separate the process of discovering the infection subgraph from the process of estimating the source. In DeepTrace, the forward component grows the observed graph, while the backward component predicts the most likely superspreader using a GNN. However, these two parts are still only loosely coupled: the forward tracing rule is not explicitly optimized based on the backward prediction quality.

This matters because the best node to expand next is not always the next node in BFS or DFS order; it is the node that is expected to improve source inference the most. A better system would let the likelihood estimator directly influence which nodes are explored next. This gap is exactly where an adaptive top-k exploration policy becomes useful.

E. Limited Modeling of Temporal or Stage-Wise Growth

DeepTrace already treats contact tracing as an iterative process, but its core mechanism still relies on a static snapshot of the current observed graph at each stage. In realistic outbreaks, however, the graph evolves over time, and the quality of the inference depends on how quickly and intelligently the tracing system reacts to newly observed information. The review literature also highlights that source detection settings vary widely across works, and some methods use multiple snapshots or temporal networks instead of a single static observation.

This suggests a gap in how tracing systems handle incremental evidence. A strong framework should not only predict the source from a snapshot, but should also adapt its exploration policy as new nodes are discovered.

F. Benchmarks and Generalization Still Need Stronger Validation

The review literature emphasizes that source detection studies often use different problem settings, epidemic assumptions, and evaluation protocols, which makes direct comparison difficult. It also notes that many GNN-based methods claim strong generalization, but those claims are still constrained by the simulation setup used to generate the training data. In DeepTrace, training is done on synthetic epidemic networks and then fine-tuned on selected high-quality labels, which is sensible, but it still leaves open the question of how robust the method is under different graph structures, edge densities, and tracing horizons.

This gap is important for this project because any proposed improvement should be tested not only on accuracy, but also on how well it transfers across graph families and tracing conditions. A better evaluation should include heterogeneous network types and compare the effect of the new features, edge weighting, and adaptive exploration separately.

In summary, the literature shows that GNNs are promising for epidemic source detection, but current methods still rely on limited node features, uniform edge treatment, and fixed exploration policies. DeepTrace improves on classical tracing by combining forward and backward inference, but it still leaves room for more adaptive and expressive design choices. The proposed DeepTrace++ framework addresses these gaps by enriching node features, incorporating edge weights, and replacing static traversal with a learned top-k expansion policy.

V. Proposed Approach

In this work, we propose DeepTrace++, an extension of the DeepTrace framework that enhances epidemic source detection by improving node representations, incorporating edge heterogeneity, and introducing an adaptive exploration strategy. The proposed method preserves the original formulation of contact tracing as a maximum-likelihood estimation problem on an incrementally observed epidemic graph.

A. Overview of the Framework

DeepTrace++ operates on an evolving contact tracing graph $G_n \subseteq G$, where n denotes the number of observed infected nodes at a given stage. The framework consists of three tightly coupled components:

1) **Feature Enrichment Module** — constructs expressive node features,
2) **Weighted Graph Neural Network (GNN)** — estimates source likelihoods,
3) **Adaptive Forward Tracing Policy** — guides graph expansion using model predictions.

At each iteration, the algorithm updates node representations, computes likelihood scores for all nodes in G_n, and selectively expands the graph based on these scores. This process continues until the tracing budget is exhausted or convergence is achieved.

B. Feature Enrichment Module

The effectiveness of GNN-based source detection depends heavily on the quality of node features. While the original DeepTrace framework uses basic features such as infected proportion and boundary distance ratio, these features primarily capture local structural information.

To improve representational capacity, we augment each node v with additional global and positional features:

- **Infected Proportion** $\hat{r}(v)$: fraction of infected neighbors,
- **Boundary Distance Ratio** $\check{r}(v)$: normalized distance to the boundary of the infected subgraph,
- **Closeness Centrality** $C(v)$: measures how close a node is to all other nodes in G_n,
- **Distance from Index Case** $d(v, v_{\text{index}})$: shortest path distance from the initial observed node.

The resulting feature vector is:

$$h_v^{(0)} = [1, \hat{r}(v), \check{r}(v), C(v), d(v, v_{\text{index}})]^T \tag{26}$$

This enriched feature space allows the model to capture both:

- **Local infection dynamics** (via infected ratio),
- **Global structural importance** (via centrality),
- **Temporal/spatial progression** (via distance from index).

C. Weighted Message Passing in GNN

In the original DeepTrace framework, message passing treats all edges equally. However, real-world interactions are heterogeneous, and not all contacts contribute equally to disease transmission.

To address this, we introduce **edge-weighted message passing**, where each edge (u, v) is associated with a weight w_{uv} representing interaction strength or transmission likelihood.

The GNN update rule is modified as:

$$h_v^{(l)} = \sigma\left(\sum_{u \in \mathcal{N}(v)} w_{uv} \cdot h_u^{(l-1)} \cdot W^{(l)}\right) \tag{27}$$

where:

- $h_v^{(l)}$ is the node embedding at layer l,
- $W^{(l)}$ are learnable parameters,
- $\sigma(\cdot)$ is a non-linear activation function.

This modification enables the model to:

- Prioritize stronger interactions,
- Suppress weak or noisy connections,
- Better approximate real-world transmission patterns.

Edge weights can be defined based on:

- Synthetic interaction strengths,
- Frequency of contact,
- Or uniform weights as a baseline.

D. GNN-Based Likelihood Estimation

The goal of the GNN is to approximate the likelihood function:

$$P(G_n \mid v) \tag{28}$$

for each node $v \in G_n$.

Instead of computing this likelihood explicitly (which is computationally expensive due to combinatorial permutations), the GNN learns to predict:

$$\hat{y}_v \approx \log P(G_n \mid v) \tag{29}$$

The node with the highest predicted likelihood is selected as the estimated source:

$$\hat{v} = \arg\max_{v \in G_n} \hat{y}_v \tag{30}$$

The model is trained using supervised learning on synthetic epidemic graphs, where the ground truth source is known. This follows the same two-phase training strategy used in DeepTrace, combining approximate and exact likelihood labels.

E. Adaptive Forward Tracing Policy

A key limitation of DeepTrace is the use of static traversal strategies (BFS/DFS) for graph expansion. These strategies do not utilize the likelihood predictions generated by the model.

To overcome this, we propose a **top-k adaptive tracing policy**, defined as follows:

At iteration t:

1) Compute likelihood scores $\hat{y}_v$ for all nodes in G_n,
2) Select the top-k nodes with highest scores:

$$S_k = \text{Top-}k\left(\{\hat{y}_v \mid v \in G_n\}\right) \tag{31}$$

3) Expand the graph by exploring neighbors of nodes in S_k,
4) Update G_n and repeat.

This approach ensures that:

- Exploration is focused on the most promising regions,
- Irrelevant parts of the graph are avoided,
- The tracing process aligns with the MLE objective.

This effectively transforms contact tracing into a **model-guided exploration problem**, where inference directly influences data collection.

F. Algorithmic Flow

The overall DeepTrace++ algorithm proceeds as follows:

1) Initialize G_1 with the index case,
2) Compute node features for all nodes in G_n,
3) Apply the weighted GNN to obtain likelihood scores,
4) Select top-k nodes based on predicted likelihood,
5) Expand the graph using selected nodes,
6) Repeat steps 2–5 until termination.

VI. Datasets

We use synthetic epidemic networks generated using standard graph models:

- Erdős–Rényi (ER) graphs,
- Barabási–Albert (BA) graphs,
- Watts–Strogatz (WS) graphs,
- Regular graphs.

Epidemic spread is simulated using the Susceptible-Infected (SI) model.

Training data consists of:

- Graph structure,
- Node states (infected/susceptible),
- Ground truth source node.

This simulation-based approach is widely used in prior works.

Later, we use real-world COVID-19 Taiwan and Hong Kong datasets, made available in the GitHub repository for the original DeepTrace paper.

VII. Evaluation Metrics

Evaluating epidemic source detection models is inherently challenging due to the stochastic nature of contagion processes and the ambiguity in identifying a unique source, especially in graphs with cycles. A single metric such as accuracy is often insufficient to fully capture model performance. Therefore, we employ a combination of ranking-based, distance-based, and tracing-based evaluation metrics to obtain a comprehensive assessment of the proposed framework.

A. Top-k Accuracy

Top-k accuracy measures whether the true source node appears within the top-k nodes ranked by the model's predicted likelihood scores.

Formally,

$$\text{Top-}k = \frac{1}{M}\sum_{i=1}^{M} \mathbf{1}\left(v_i^* \in \hat{S}_k^{(i)}\right) \tag{32}$$

where:

- M is the number of test samples,
- v_i^* is the true source node,
- $\hat{S}_k^{(i)}$ represents the top-k predicted nodes,
- $\mathbf{1}(\cdot)$ is the indicator function.

This metric reflects how effectively the model prioritizes the correct infection source among the most likely candidates.

B. Top-1 Accuracy

Top-1 accuracy represents the fraction of cases where the model correctly identifies the exact source. This is the strictest evaluation metric and reflects the model's ability to precisely recover the true origin of the epidemic. However, due to the inherent uncertainty in epidemic spreading, this metric alone can be overly harsh. For example, in highly connected graphs, several nodes may produce identical infection patterns, making exact recovery theoretically impossible beyond a certain time horizon.

Thus, Top-1 accuracy is reported alongside other metrics rather than being used as the sole evaluation criterion.

C. Normalized Rank of the True Source

To better evaluate the ranking quality of predictions, we use the normalized rank metric:

$$R = 1 - \frac{1}{N}\sum_{i=1}^{N} \text{rank}(v^*) \tag{33}$$

where $\text{rank}(v^*)$ denotes the position of the true source in the sorted list of predicted likelihoods, and N is the number of nodes.

This metric provides a continuous measure of how well the model prioritizes the correct node, even when it is not ranked first. It is particularly useful in scenarios where multiple nodes have similar likelihood scores. Prior work highlights that ranking-based metrics offer deeper insight into model performance compared to binary accuracy measures, especially under uncertainty.

D. Average Error Distance

The average error distance measures how far the predicted source is from the true source in terms of graph distance:

$$\text{Average Error} = \frac{1}{M} \sum_{i=1}^{M} \text{dist}(v_{\text{pred}}, v^*) \tag{34}$$

where M is the number of test instances.

This metric captures the spatial error of the prediction. Even if the model fails to identify the exact source, a prediction that is close to the true source in the graph is still valuable. This is especially relevant in contact tracing, where identifying the correct region or cluster of infection may be sufficient for intervention.

DeepTrace itself uses this metric to evaluate how quickly the predicted source converges toward the true source during the tracing process.

E. First Detection Time

First detection time measures how early in the tracing process the model correctly identifies the true source:

$$\text{First Detection Time} = \min\{t \mid v_t = v^*\} \tag{35}$$

This metric is specific to iterative tracing frameworks such as DeepTrace, where the observed graph grows over time. It evaluates how efficiently the algorithm converges to the correct solution as more nodes are explored.

A lower first detection time indicates that the model can quickly focus on the most relevant parts of the graph, which is crucial in real-world outbreak scenarios where early intervention is essential. This metric directly reflects the effectiveness of the forward tracing strategy and is therefore particularly important for evaluating the proposed adaptive top-k exploration mechanism.

F. Coverage Efficiency

Coverage efficiency evaluates how much of the graph must be explored before correctly identifying the infection source:

$$\text{Coverage Efficiency} = \frac{|V_{\text{explored before detection}}|}{|V_{\text{total}}|} \tag{36}$$

Lower values indicate more efficient tracing, since fewer nodes need to be explored before detecting the source.

G. Reason for Multiple Metrics

No single metric fully captures the complexity of the source detection problem. Specifically:

- Top-1 accuracy measures exact correctness,
- Top-k accuracy and normalized rank evaluate ranking quality,
- Average error distance measures spatial closeness,
- First detection time and efficiency evaluate tracing performance.

Together, these metrics provide a holistic evaluation of both **prediction quality** and **exploration efficiency**, which is essential for assessing iterative contact tracing frameworks like DeepTrace and its proposed extension.

VIII. Results

We compare four model variants: DeepTrace (baseline), DeepTrace++ Enriched, DeepTrace++ Weighted, and the full DeepTrace++ model. The baseline uses the original GraphSAGE architecture with 8 structural node features and uniform edge treatment. The Enriched variant adds closeness centrality and normalized distance from the index case (10 features total), while the Weighted variant introduces degree-based edge-weighted message passing without changing features. We evaluate the models across five synthetic network topologies, each with around 200 nodes: Erdős–Rényi (ER), Barabási–Albert (BA), Watts–Strogatz (WS), Regular, and Stochastic Block Model (SBM) along with the real world Hong Kong and Taiwan COVID-19 data made available by the original DeepTrace authors. Each setting is evaluated over 20 trials, with 150 epochs each. Metrics include Top-k Accuracy, Normalized Rank, Average Error Distance, First Detection Time, and Coverage Efficiency.

A. Overall Performance

Table I summarizes performance averaged across all topologies.

TABLE I
Overall Performance Across All Topologies

Model	Top-1	Top-5	Top-10	Error	Detect	Cov. Eff.
DT++ enriched	0.62	0.95	1.00	0.61	98.68	0.987
DeepTrace	0.61	0.93	0.99	0.68	98.84	0.988
DeepTrace++	0.58	0.95	1.00	0.66	98.84	0.988
DT++ weighted	0.50	0.92	1.00	0.86	99.16	0.992

B. Per-Topology Performance

TABLE II
Performance on ER Networks

Model	Top-1	Top-5	Error	Time (s)
DT++ enriched	1.00	1.00	0.00	0.118
DT++ weighted	1.00	1.00	0.00	0.106
DeepTrace	1.00	1.00	0.00	0.118
DeepTrace++	1.00	1.00	0.00	0.115

TABLE III
Performance on BA Networks

Model	Top-1	Top-5	Error	Time (s)
DT++ enriched	0.45	0.80	0.90	0.104
DT++ weighted	0.40	0.80	0.95	0.090
DeepTrace	0.45	0.80	0.90	0.098
DeepTrace++	0.40	0.80	0.95	0.107

TABLE IV
PERFORMANCE ON WS NETWORKS

Model	Top-1	Top-5	Error	Time (s)
DT++ enriched	0.30	0.95	1.25	0.141
DT++ weighted	0.25	0.80	1.55	0.110
DeepTrace	0.25	0.85	1.60	0.104
DeepTrace++	0.35	0.95	1.05	0.110

TABLE V
PERFORMANCE ON REGULAR NETWORKS

Model	Top-1	Top-5	Error	Time (s)
DT++ enriched	0.85	1.00	0.15	0.122
DT++ weighted	0.50	1.00	0.80	0.104
DeepTrace	0.85	1.00	0.15	0.120
DeepTrace++	0.75	1.00	0.35	0.117

TABLE VI
PERFORMANCE ON SBM NETWORKS

Model	Top-1	Top-5	Error	Time (s)
DT++ enriched	0.50	1.00	0.75	0.132
DT++ weighted	0.35	1.00	1.00	0.114
DeepTrace	0.50	1.00	0.75	0.114
DeepTrace++	0.40	1.00	0.95	0.120

C. Feature Ablation Study

TABLE VII
EFFECT OF FEATURE ENRICHMENT

Setting	Top-1	Mean Rank	Time (s)
Original Features	0.64	0.1067	83.99
Enriched Features	0.67	0.1033	96.90

D. Edge Weighting Study

To evaluate the effect of interaction strength modeling, we consider three edge-weighting schemes:

- **Weighted-degree:**
  Edge weights are proportional to the degree of neighboring nodes:

$$w_{uv} = \frac{\deg(u) + \deg(v)}{2} \tag{37}$$

- **Weighted-uniform:**
  All edges are assigned identical weights:

$$w_{uv} = 1 \tag{38}$$

- **Weighted-random:**
  Edge weights are sampled from a uniform distribution:

$$w_{uv} \sim \mathcal{U}(0, 1) \tag{39}$$

These weighting strategies allow evaluation of how deterministic and stochastic interaction strengths influence message passing and source detection accuracy.

TABLE VIII
IMPACT OF EDGE WEIGHTING

Scheme	Top-1	Mean Rank	Time (s)
Unweighted	0.62	0.1100	24.76
Weighted-degree	0.64	0.1067	24.67
Weighted-uniform	0.66	0.1000	25.26
Weighted-random	0.61	0.1100	25.46

E. Evaluation on Real-World Datasets

To assess generalization beyond synthetic graphs, we evaluate DeepTrace and DeepTrace++ on real-world epidemic contact tracing datasets derived from Hong Kong and Taiwan. The datasets include BFS- and DFS-based partial tracing trees as well as real infection trees.

TABLE IX
PERFORMANCE ON (HONG KONG)BFS_PART_TREE_10.CSV

Model	Top-1	Top-5	Top-10	Hop Error	Detect	Time (s)
DeepTrace	0	0	0	3	43	0.0060
DT++ enriched	0	0	0	3	43	0.0064
DT++ weighted	0	0	0	3	43	0.0058
DeepTrace++	0	0	0	3	43	0.0060

TABLE X
PERFORMANCE ON (HONG KONG)DFS_PART_TREE_10.CSV

Model	Top-1	Top-5	Top-10	Hop Error	Detect	Time (s)
DeepTrace	0	0	0	3	43	0.0060
DT++ enriched	0	0	0	3	43	0.0059
DT++ weighted	0	0	0	3	43	0.0063
DeepTrace++	0	0	0	3	43	0.0057

TABLE XI
PERFORMANCE ON TW_DATA_43_TREE.CSV

Model	Top-1	Top-5	Top-10	Hop Error	Detect	Time (s)
DeepTrace	1	1	1	0	20	0.0032
DT++ enriched	1	1	1	0	20	0.0031
DT++ weighted	1	1	1	0	20	0.0033
DeepTrace++	1	1	1	0	20	0.0031

TABLE XII
PERFORMANCE ON TW_DATA_76_TREE.CSV

Model	Top-1	Top-5	Top-10	Hop Error	Detect	Time (s)
DeepTrace	1	1	1	0	38	0.0043
DT++ enriched	1	1	1	0	38	0.0051
DT++ weighted	1	1	1	0	38	0.0044
DeepTrace++	1	1	1	0	38	0.0043

TABLE XIII
AVERAGE PERFORMANCE ACROSS REAL-WORLD DATASETS

Model	Top-1	Top-5	Top-10	Hop Error	Detect	Time (s)
DT++ enriched	0.50	0.50	0.50	1.50	36.0	0.00513
DT++ weighted	0.50	0.50	0.50	1.50	36.0	0.00494
DeepTrace	0.50	0.50	0.50	1.50	36.0	0.00488
DeepTrace++	0.50	0.50	0.50	1.50	36.0	0.00480

1) Overall Real-World Performance:

2) Discussion: The results reveal a clear contrast between partial tracing datasets and complete infection trees. On the partial BFS and DFS datasets, all models fail to correctly identify the source (Top-1 = 0), with a consistent hop error of 3. This indicates that limited observability significantly degrades source inference performance, regardless of model sophistication.

In contrast, on the Taiwan datasets representing more complete infection trees, all models achieve perfect accuracy (Top-1 = 1.0) with zero hop error. This highlights that when sufficient structural information is available, both DeepTrace and DeepTrace++ can reliably recover the true source.

Overall, the average Top-1 accuracy of 0.50 reflects this bimodal behavior. The results suggest that the primary bottleneck in real-world performance is not model capacity, but the completeness of the observed contact tracing graph. Improvements in adaptive exploration or partial observability handling are therefore critical directions for future work.

F. Key Observations

- DeepTrace++ (enriched) consistently improves Top-1 Accuracy and reduces error distance compared to DeepTrace.
- Performance varies significantly across graph structures, with BA and WS being the most challenging.
- Feature enrichment improves ranking quality (lower mean rank).
- Edge weighting provides modest gains, with uniform weighting performing best.

IX. Conclusion

In this work, we proposed *DeepTrace++*, an enhanced graph neural network framework for epidemic source detection that builds upon the original DeepTrace formulation. By incorporating centrality-aware node features, edge-weighted message passing, and improved structural representations, the proposed approach aims to better capture both local and global properties of epidemic diffusion processes.

Extensive experiments on synthetic network topologies demonstrate that DeepTrace++ consistently improves Top-1 accuracy and reduces average error distance compared to the baseline DeepTrace model, particularly in complex graph structures such as Barabási–Albert and Watts–Strogatz networks. The enriched feature variant achieves the best overall performance among the learned models, highlighting the importance of incorporating richer structural information into the learning process. Furthermore, ablation studies confirm that both feature augmentation and edge-weighting strategies contribute to improved ranking quality, with measurable gains in normalized source rank.

Evaluation on real-world Hong Kong and Taiwan epidemic datasets shows that all models, including DeepTrace++, achieve perfect source identification when the infection graph is fully observed. This indicates that under ideal conditions, the source detection problem becomes well-defined and can be solved reliably by both learning-based and classical approaches. However, the contrast with synthetic experiments suggests that the primary challenge lies in handling incomplete or partially observed networks, where structural ambiguity makes inference significantly harder.

Overall, the results highlight that while DeepTrace++ improves inference quality in challenging settings, the effectiveness of source detection is fundamentally influenced by the completeness and quality of observed contact tracing data. This underscores the importance of integrating adaptive exploration strategies with learning-based inference.

Future work can focus on extending DeepTrace++ to explicitly handle partial observability, incorporating probabilistic or temporal infection models, and developing fully adaptive forward tracing policies guided by model uncertainty. Additionally, scaling the framework to larger real-world contact networks and integrating real-time data streams remain important directions for practical deployment.

In summary, DeepTrace++ advances the state of graph-based epidemic source detection by improving both representation learning and inference robustness, providing a more flexible and effective framework for real-world contact tracing applications.

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